

OUSSAMA EL-ASRI

Full-Stack AI / GenAI Engineer · Software Engineering Student · Seeking 4–6 Month Internship

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SUMMARY

Self-directed Software Engineering student at UM6P-1337 (42 Network) specializing in full-stack AI and GenAI applications — RAG pipelines, LLM-powered agents, backend APIs, and containerized multi-service systems. Strong foundation in systems programming, networking, parsing, and software architecture. Accustomed to solving technical problems independently through documentation, research, and project-based development.

TECHNICAL SKILLS

Languages: Python, C, C++98, JavaScript, TypeScript
AI / GenAI: RAG, LangChain, LangGraph, CrewAI, Ollama, AI Agents, vector embeddings, prompt engineering
Frontend: React, Next.js, Tailwind CSS, Vite
Backend: FastAPI, Flask, Django, Express.js, REST APIs, Gunicorn, Uvicorn
Databases: PostgreSQL, SQLite, Redis, Prisma ORM, SQLAlchemy
DevOps: Docker, Docker Compose, NGINX, Git, Linux, uv
Systems: HTTP/1.1, TCP/IP, CGI, non-blocking I/O, concurrency, mutexes/semaphores, process management

SELECTED PROJECTS

AI Studio — Unified AI Platform *Personal*

React · TypeScript · Vite · Python · Flask · FastAPI · LangChain · LangGraph · Ollama · Docker Compose

- Built a full-stack AI platform integrating two independent AI services behind a unified React interface.
- Developed a RAG assistant that ingests arbitrary URLs, chunks content, generates vector embeddings with nomic-embed-text, and performs semantic retrieval for contextual Q&A.
- Built a Crypto Analysis Agent with LangChain and LangGraph to process natural-language requests, retrieve 30-day OHLCV data from Binance, and calculate EMA, RSI, OBV, standard deviation, and Bollinger Bands.
- Designed a multi-container architecture with separate Flask/FastAPI services sharing a local Ollama LLM runtime.

ft_transcendence — Real-Time Multiplayer Platform *Team project, 4 members*

Next.js · Express.js · Socket.io · Prisma · SQLite · Python · Flask · LangChain · Ollama · Docker

- Owned the AI subsystem of a full-stack real-time multiplayer platform.
- Designed and implemented a RAG-powered AI assistant using LangChain and Ollama, including content retrieval, embeddings, prompt orchestration, and backend integration.
- Integrated the AI service into a broader platform featuring real-time multiplayer gaming, WebSocket chat, JWT/OAuth authentication, 2FA, statistics, and analytics.
- Worked within a modular multi-service architecture orchestrated with Docker Compose and NGINX.

Webserv — HTTP/1.1 Web Server *Team project*

C++98 · poll() · CGI · Python · PHP-CGI

- Implemented HTTP response generation for GET, POST, and DELETE — status codes, headers, payloads, file uploads, and error responses.
- Built Python CGI execution: process management, environment setup, request-body handling, and output parsing.
- Contributed to a non-blocking HTTP/1.1 server architecture using a single poll() event loop for concurrent connections.

Additional Projects: **JSON Parser** — Python recursive-descent parser, custom lexer, AST construction · **Minishell** — C Unix shell: parsing, pipelines, redirections, built-ins · **Weather App** — React + FastAPI + PostgreSQL/SQLModel · **Inception** — Docker Compose: NGINX, WordPress/PHP-FPM, MariaDB, TLS

EDUCATION

UM6P-1337 — 42 Network, Morocco | 2023 – Present

Software Engineering Student · Expected Bachelor's completion after internship. Project-based, peer-to-peer curriculum covering software engineering, algorithms, systems programming, networking, DevOps, and AI.

FST Al Hoceima | 2020 – 2022

MIP — Mathematics, Computer Science & Physics

Baccalaureate — Physical Sciences · 2019

CERTIFICATIONS & ADDITIONAL

AI / Software: Anthropic — AI Fluency: Framework and Foundations · IBM — Applied Software Engineering Fundamentals · IBM — Python for Data Science, AI & Development · Vercel — React Foundations for Next.js · University of Leeds — Foundations of Software Testing and Validation

Languages: Arabic — Native · English — B2 (Upper-Intermediate) · French — Conversational

Competitive Programming: LeetCode · CodinGame · HackerRank